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A probability distribution of dependencies in interlanguage

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Abstract: The diversity of syntactic units in second language has attracted much scholarly attention. Most existing studies on syntactic diversity have focused on only a small number of syntactic structures, and it is difficult to find studies that consider the full range of syntactic dependencies present in the dataset. Based on a syntactic annotated interlanguage corpus that we constructed, this paper is a quantitative study of dependencies in English-speaking Chinese learners' interlanguage across proficiency levels. We fit the frequency distributions of dependency type, word class (both as dependent and governor), verb as a governor, and noun as a dependent with a modified right-truncated Zipf-Alekseev distribution and Zipf's law. Our findings show that: (1) from the mathematical model, interlanguage followed distributional regularities like natural languages in terms of the syntactic structure distribution; (2) most of the determination coefficients' R^2 were high, indicating that the investigated distributions in interlanguage fit the distributional law finding in natural languages. This also demonstrated that both interlanguages and natural languages consistently conform to the law of linguistic diversity and uniformity; (3) the dependency relation distribution parameters a and b manifest the developmental trend of L2 learners' proficiency levels, demonstrating that the parameters had universal applicability in reflecting interlanguage proficiency.

Keywords: Chinese interlanguage; distribution; quantitative linguistics; syntactic diversity

1 Introduction

Research employing a quantitative corpus linguistics (QCL) methodology has a long heritage (Durrant et al. 2021). Numerous previous language development studies

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have focused on the use and distribution frequency of linguistic elements and linguistic structures (mainly specific language phenomena) of language learners. The important role of the traditional QCL is to provide information on linguistic structures' frequencies. Corpus Linguistics focuses on frequencies, which can provide the language researcher with language evidence (Gries 2008).

1.1 A brief introduction to quantitative corpus linguistics

Divjak (2019: 30) states “Zipf, Yule and Herdan are three of the first and most well-known researchers in the area of word frequency distributions from the Anglo-Saxon tradition”. They proposed and verified a number of language laws, such as functional law and developmental law. Additionally, they introduced effective mathematical and statistical methods into linguistics, laying a solid foundation for the emergence of quantitative linguistics (Liu 2017).

Altmann inherited the achievements of quantitative linguistics research and developed them into the Altmann School which thrived in Central Europe in the 1970s (Köhler 2012: 7; Yih 2021). The Altmann school promoted the development of quantitative linguistics and contributed to the birth of modern quantitative linguistics. Modern quantitative linguistics is a data-intensive field, which covers various linguistic phenomena and their relationships in real language communication activities. The authentic language materials are precisely measured, observed, simulated, modeled, and explained by mathematical quantitative methods (as opposed to mathematical qualitative methods such as algebra) such as probability theory, stochastic equations, differential equations, and functional theory. The aim of modern quantitative linguistics is to uncover the mathematical laws and cooperative relations behind language phenomena, reveal the internal causes of various language phenomena, and explore the adaptive mechanism of language systems and the motivation behind language evolution (Altmann 2005; Best 2006; Liu and Huang 2012). In addition to exploring universal linguistic laws, quantitative linguists also observe the linguistic diversity common to natural languages and the internal individual features of a language through metrics derived from the laws of linguistics (Köhler 2012). In the modern quantitative linguistics research paradigm, computers are used to extract data from corpora for precise study of the structure and development of language. In sum, from the perspective of the research object, one of the strengths of quantitative linguistics is that it is concerned with the analysis of authentic, real-world texts; in terms of the research method, quantitative linguistics is precise; and regarding the research aim, quantitative linguistics is dynamic (Liu 2017). Quantitative linguistics in this paper refers to the modern quantitative

linguistics of the Altman school, and the research method in this paper is also follows that of Altman school.

Zipf was one of the founders of quantitative linguistics who espoused the frequency-based approach to linguistic studies. Zipf (1949) first abstracted universal linguistic laws from the rank-frequency regularities of texts. He found that the frequencies of words in natural texts were unequal, and that there is a law between ordinal number and frequency, which can be fitted with a power function; the resulting distribution pattern of words is called the power law distribution. All human languages follow this law.

Zipf abstracted universal linguistic laws from rank-frequency relations, and conducted considerable statistical analysis and validation work. In honor of Zipf's work and the foundation he laid for the quantitative linguistics, the power law distribution of language has come to be known as Zipf's law. The power law distribution is one of the earliest statistical laws in quantitative linguistics. It has also been demonstrated that many other linguistic units' frequency distributions in natural languages also have rank-frequency relationships, following the power law distribution. This includes dependency distance in natural languages (Liu 2007), syntactic dependency properties (Čech et al. 2018), semantic properties (polysemy and synonymy) of language (Čech et al. 2017; Gao et al. 2014), and the valency pattern of verbs (Köhler et al. 2005; Liu 2011).

Zipf regards language as a natural phenomenon, so he follows the natural sciences' observation-data-hypothesis-explanation and verification-theorem research methodology (Liu 2017). Since the discovery of the power-law distribution of words in 1935, Zipf had been searching for its cause from the perspective of linguistics. In 1949, Zipf proposed the principle of the least effort to explain this law. This principle results from two economic principles that are in competition with each other—the force of unification and the force of diversification. Language communication involves both the speaker and the addressee. From the speaker's perspective, the fewer words they produce, the more economical they are. Consequently, there will be a lot of polysemes. The speaker's economic force constitutes the unification force. However, analyzed from the addressee, to comprehend with the least effort, the morphology should be one-to-one correspondence with its semanteme, with each word carrying only one distinct meaning being preferable. In short, more words might be required in the inventory, giving rise to the diversification force (Zhang and Liu 2017). Only when uniformity and diversity achieve a balanced state of dynamic equilibrium can both sides in the communication achieve efficiency (Liu 2017). This balance is the simultaneous minimization of both efforts, which optimizes the cost of communicative transactions between the two parties. With both forces of unification and its opposite, diversification (or collectively called Zipfian processes) in play,

language evolves (Zhang and Liu 2017). Zipf's law indicates a consequence of reaching a balance between the previous two opposing forces. In fact, this leads to a high frequency of use for a tiny number of words and a low frequency of use of a large number of words. The word frequency distribution curve has a long tail, and the rank frequency distribution of low-frequency words is reflected in it, representing the diversity of power distribution.

The Zipf-Alekseev model is a variant of Zipf's Law (the power law distribution) (Popescu et al. 2014) which also follows the principle that language units are diverse. It has been applied to the distribution of word frequency, syntactic relations, and discourse relations (Hřebíček 2002; Liu 2009b; Yue and Liu 2011). Liu (2009b) fit the Zipf-Alekseev model with the distributions of Chinese syntactic dependency properties, and found that most of the investigated distributions fit well. This demonstrated that there was a quantitative linguistic law in the distribution of Chinese linguistic units' diversity. In short, previous quantitative studies have found that Zipf's Law not only applies to many types of natural languages, but that language units also follow a rank-frequency distribution. The rank-frequency distribution among the functions related to Zipf's law can "characterize a diversification process" (Pan and Liu 2014: 2). In other words, provided that the rank frequency distribution of a language structure conforms to Zipf's law, it can prove the existence of the principle of least effort, and reflect the diversity of the language structure.

1.2 The application of quantitative corpus linguistics in second language acquisition

Interlanguage is a special natural language produced by second language learners (henceforth L2) when they are acquiring a new language. Interlanguage has a structurally intermediate status between the native and target languages (Selinker 1972), and has the dynamic features of continuous and progressive development (Corder 1967; Nemser 1971; Selinker 1972). Additionally, interlanguage has permeability (Adjemian 1976). Interlanguage is influenced by both the native language (NL) and the target language (TL). NL and TL are the two basic sources of interlanguage. The linguistic rules of NL can penetrate the interlanguage system through language transfer, and at the same time, the TL penetrates the interlanguage system through over-extension and distortion. Consequently, we cannot simply regard the target language as a language hybrid formed by the interference of NL and TL. Interlanguage has its own phonological, syntactic and lexical rules, showing the feature of independence from both native language and target language (Adjemian 1976; Selinker 1972). In addition, throughout the process of language development, L2

learners' interlanguage system is in flux. L2 learners are constantly revising, enriching and improving their language rules. However, this developmental process is not linear, but complex and dynamic (Corder 1967; Larsen-Freeman 1997; Nemser 1971; Selinker 1972).

One of the most important topics in the field of second language acquisition (henceforth SLA) is identifying metrics that can measure L2 proficiency. Many scholars have argued that syntactic complexity can measure L2 proficiency, and can reflect the complexity and diversity of interlanguage (Ellis 2003, 2009; Housen and Kuiken 2009). Put differently, syntactic complexity covers two dimensions: syntactic sophistication and syntactic diversity (Bulté and Housen 2014; Lu 2011; Ortega 2003, 2012). However, most studies which investigate syntactic complexity in interlanguage have focused on the former. Research on the syntactic diversity of second language has been limited (Bi and Jiang 2020) and should not be neglected (Bi and Jiang 2020; De Clercq and Housen 2017; Park 2017). Most studies of the development of syntactic diversity in interlanguage have used partial or specific linguistic units as metrics. These have included the predicate-argument structure, the noun phrase, word combinations and phrasal logical units as metrics (Ellis et al. 2015; Kyle and Crossley 2018; Paquot 2018; Park 2017; Zhao and Jiang 2020). For example, Zhao and Jiang (2020) found that the distributional features of verb valency in interlanguage had regular probability distributions. Moreover, parameter a in Zipf's law decreased along with L2 proficiency level while parameter b increased. This demonstrated that the verb valency parameters a and b in Zipf's law, to an extent, can manifest in L2 proficiency levels. Yet current research on second language syntactic diversity remains insufficient, and there is still immense growth potential for the use of syntactic diversity indices to measure L2 proficiency.

Considering the research content, most previous researchers have investigated the partial linguistic structure's diversity of interlanguage. For example, Zhao and Jiang (2020) only investigated the distribution features of various verb valency structures in English as an interlanguage with a mathematical model which analysed the development of interlanguage verbs' distributional properties. It did not explore the syntactic diversity of the entire interlanguage system, i.e., it does not examine the distribution of learners' overall syntactic structures. Moreover, few studies have investigated the features of interlanguage using mathematical models from the perspective of quantitative linguistics. Even fewer have investigated the overall syntactic structure distributions in interlanguage. Furthermore, determining whether laws of quantitative linguistics also apply to interlanguage requires expanding the quantitative linguistics sample. In other words, various language samples must be used to test the general applicability of the laws of quantitative linguistics. Additionally, concerning the research object, existing quantitative

linguistic research on second language has focused on English as interlanguage. Other languages have received scant attention. The current conclusions are in need of further verification using other languages as interlanguage. This would help SLA researchers probe into metrics that could measure the syntactic complexity of interlanguage and overall language development.

1.3 The present study

From the perspective of linguistic morphology typology, Chinese, representing an isolating language, differs from the inflectional languages, such as English, in which the shapes of words are altered to express grammatical information such as tense, number, and case. Most English words belong to one and only one part of speech in sentences. However, the syntactic function of Chinese word classes enjoys greater freedom. As an isolating language, Chinese uses little or no inflection to indicate grammatical relationships. Moreover, the correspondence between Chinese word types and syntactic relations is more complex (Zhu 1985). This begs the question: can existing conclusions be verified in Chinese as interlanguage? This requires further investigation.

Recent progress in computer science and natural language processing have provided the opportunity to employ a new quantitative linguistics approach. It is now possible to investigate syntactic diversity into interlanguage using a suitable mathematical model. Additionally, we can now also explore the universal law of natural languages from a holistic perspective. Dependency grammar-based syntactic corpora provide support for the quantitative analysis of syntax. They also enable the quantification and measurement of syntactic units, which can improve the precision of syntactic studies (Hudson 2007; Köhler 2012; Ninio 2006; Tesnière 1959). Furthermore, language learners, especially beginners, often produce ungrammatical constructions. It has been shown that dependency grammar is more suitable than traditional phrase structure grammars for analysing learners' deep syntactic structure relationships (Kyle 2016). According to the principles of dependency grammar, a sentence is an organized and complete structural unit whose basic components are words. Sentence structures are determined by the dependency relations between syntactic units. A dependency relation is composed of two non-symmetric directed structural elements: the governor and the dependent (Hudson 2007; Liu 2009a; Mel'čuk 2003). The dependency type can identify grammatical structures, which can systemically extract a complete object set of syntactic information and various syntactic metrics (Meurers and Dickinson 2017). As shown in Figure 1, the dependency types SUBJ,

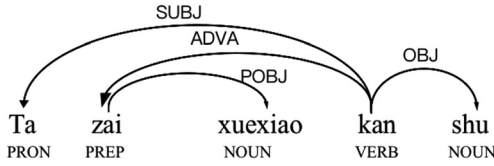


Figure 1: Dependency structure in a Chinese sample sentence: *Ta zai xuexiao kan shu* 'He reads books at school'.

OBJ, ADVA, and POBJ represent subject-predicate, verb-object, adverbial-predicate and preposition-object structures, respectively. More generally, the number of dependency relations in a text represents the total number of grammatical relations in the text, and the type of dependency relations equals the total number of grammatical relation types (Bi and Jiang 2020). Mohanty and Popescu (2014: 97) state “every linguistic entity and all its properties underlie evolution and diversification in any possible sense”. Therefore, the overall statistics and fitting data of tokens and types to the three important properties of the dependency (governor, dependent and dependency type) facilitate the analysis of the distribution and diversification of the syntactic structures of a text or a corpus. This is an effective method for studying syntactic diversity. Because of interlanguage’s natural linguistic properties (Selinker 1972), findings from quantitative linguistics also shed light on the diversity of syntactic development in interlanguage. Previous studies have also demonstrated this method’s effectiveness (Bi and Jiang 2020; Chen and Xu 2019; Eppler 2014; Ouyang and Jiang 2018; Zhao and Jiang 2020).

Based on Chinese syntactically annotated corpora, the present study employs the quantitative linguistics approach, taking advantage of the quantitative properties of the dependency relation. In this way, it serves as a quantitative study of the syntactic diversity of Chinese interlanguage produced by native English speakers with varying language competencies. This article addresses the following questions:

- (1) Do the distributions of Chinese interlanguage’s syntactic dependency properties follow those distributional regularities which have been identified in the natural languages?
- (2) Do the distributions of Chinese interlanguage’s syntactic dependency properties of the core governor (verb) and dependent (noun) follow the same distributional regularities as the natural languages?
- (3) If Chinese interlanguage conforms to Zipf’s law, are parameters in the corresponding law relevant to language proficiency, and can they serve as universal metrics to reflect L2 proficiency levels?

2 Materials and methods

2.1 Participants and materials

We built a syntactic corpus containing 99 compositions written by 99 English-speaking Chinese L2 (ECSL) learners who were majoring in Chinese language at a university in Beijing. The participants consisted of university students from all 4 years (freshmen, sophomores, juniors, seniors), and were aged between 20 and 25. All had learned Chinese from scratch in college. We classified the ECSL learners' compositions into four proficiency levels (G1–G4). The participants were required to write Chinese compositions within a period of 50 min on the printed answer sheet. There was only one composition title on the answer sheet, and the answer sheets with different composition titles were randomly distributed to the participants. During this period, they were not allowed to refer to any materials. After completing the written task, they agreed to have their written corpora collected for academic analysis. Every composition for each proficiency level had been written by a mid-level student from that level. The total number of participants, the year of Chinese learning, HSK (Chinese Proficiency Test) level and L2 vocabulary size for each level are shown in Table 1. HSK (from level I to VI) is an international standardized test which assesses non-native Chinese speakers' aptitudes in using the Chinese language in their daily, academic and professional lives.

The syntactically annotated corpus of Chinese interlanguage established in this study contains 99 texts with a total of 19,996 tokens and about 5,000 tokens for each level. The topic of every composition pertains to ECSL learners' daily lives. We controlled the consistency of genres and topics. All compositions are narrative with topics familiar to the L2 learners, such as "My daily life", "My hobbies", "A person I am familiar with" and "A book/movie I have read or watched".

Table 1: Participant profile.

Proficiency levels	Participants	Words	HSK level	Vocabulary size (in words)	Years of Chinese learning
G1	46	5,000	II	500	0.5–1
G2	27	5,006	III	800	1.5–2
G3	16	5,000	IV	2,000	2.5–3
G4	10	4,996	V ~ VI	4,000	3.5–4

2.2 Procedure and data analysis

The annotation process can be divided into the following procedures: word segmentation, part of speech (henceforth POS) tagging, syntactic dependency tagging and error tagging. First, we annotated the POS annotation and dependency relation automatically by Language Technology Platform (LTP) (Che et al. 2021) tagging software developed by the Research Center for Social Computing and Information Retrieval at Harbin Institute of Technology. Then, three linguistic experts conducted manual proofreading, syntactic information addition and error labeling on the basis of preliminary labeling. This was done using the dependency relation syntactic annotation system in modern Chinese as a reference (Liu and Huang 2006). This system contains 34 dependency relations. When there was disagreement, they would discuss before deciding on the sentence’s final annotation. Additionally, we did not delete L2 learners’ syntactic error dependency relations, i.e., the linguistic errors were presented in the corpus faithfully. In this way, we could present second language learners’ actual language use.

Figure 1 shows the dependency analysis of the sentence *Ta zai xuexiao kan shu* ‘He reads books at school’. The syntactic labels above the arc indicate the specific dependency relation between the governor and the dependent. To facilitate the calculation, the annotation in Figure 1 can be converted into Table 2. In Table 2, each row has a dependency pair consisting of the three important components of the syntactic dependency property—the dependent, the governor, and the dependency type. The corpus annotated by the above syntactic tagging scheme is called the treebank. The treebank contains extensive syntactic information, and therefore, the interlanguage treebank is an effective tool and resource for investigating interlanguage syntactic features.

By constructing an interlanguage treebank as shown in Table 2, we can extract all of the syntactic dependency properties from interlanguage, and thus observe the syntactic diversity of Chinese interlanguage and its linguistic laws. This paper

Table 2: Annotation of the sample Chinese sentence: *Ta zai xuexiao kan shu* ‘He reads books at school’.

Dependents			Governors			Dependency type
Order number	Words	POS	Order number	Words	POS	
1	Ta/He	r	4	Kan/read	v	SUBJ
2	Zai/in	p	4	Kan/read	v	ADVA
3	Xuexiao/school	n	2	Zai/in	p	POBJ
4	Kan/read	v				
5	Shu/books	n	4	Kan/read	v	OBJ

explores the distributions in the general dependency, governor and dependent types in Chinese interlanguage. In addition, we also consider the distributions of the dependency properties of the most frequent governor (verb) and dependent (noun). Then, using the Altmann-Fitter, we fit the mathematical models (Zipf's law and the Zipf-Alekseev model) with the investigated data (Liu 2009b). The distribution of the power law equation is:

$$y = ax^b \quad (1)$$

y is the word frequency, x is the rank, and a and b are parameters.

The Zipf-Alekseev distribution is a variant of Zipf's law, and can be expressed by Formula (2):

$$P_x = P_1 x^{-(a+b \ln x)}, x = 1, 2, 3, \dots \quad (2)$$

P_1 is the logarithm of the ratio of the probabilities, and P_x is proportional to the logarithm of the class size. If Formula (3) is considered to follow a probability distribution, then P_1 is the normalization constant; otherwise, it is the size of the first category, which is the case of $x = 1$. In most cases, diversification distributions display a diverging frequency in the first class while the rest of the distribution behaves regularly. In such cases, the first class is usually assigned a particular value, α ; then Formula (2) can be transformed into:

$$P_x = \begin{cases} \alpha, & x=1 \\ \frac{(1-\alpha)x^{-(a+b \ln x)}}{\sum_{j=2}^n j^{-(a+b \ln j)}}, & x=2, 3, \dots, (n) \end{cases} \quad (3)$$

where:

$$T = \sum_{j=2}^n j^{-(a+b \ln j)}, a, b \in \mathbb{N}, 0 < a < 1. \quad (4)$$

Formulas (3) and (4) are called the Zipf-Alekseev distribution. If n is finite, then Formula (3) is called the right truncated modified Zipf-Alekseev distribution.

3 Results and discussion

3.1 Overall dependency properties

We first report the distributions of general dependency types in Chinese interlanguage. According to the dependency grammar, all linguistic units in a sentence can form a "hierarchical system", and the relations between them can be divided into

“complements” and “adjuncts”. In Chinese, a simple sentence usually contains complements, such as *subject* (subj), *object* (obj), *predicate complement* (comp),¹ *subobject* (subobj), and *subject complement* (soc).² The adjuncts are the modifiers of the sentence, such as *adverbials* (adva), *attributives* (atr), *coordinating adjuncts* (coor), *clause adjuncts* (cr), *aspect adjuncts* (ta), *verb adjuncts* (va), and *insertion* (ina) (Liu and Huang 2006). The diversity of dependency types represents the diversity of syntactic structure use, and ECSL learners’ rank-frequency distributions of syntactic structures are shown in Figure 2. Visual inspection of Figure 2 indicates that not every syntactic structure is used with the same frequency, and the distribution of syntactic structure types of interlanguage has a long tail. This seems to present a power-law distribution in line with the principle of least effort. It also shows that there are diversified characteristics in the syntactic structures of L2 learners. Chinese learners with different language competencies were able to produce various syntactic structures, and students in all levels used over 30 dependency types. However, most dependency types were typical syntactic structures, such as *adverbials* (adva), *subjects* (subj), *objects* (obj), and *attributes* (atr). With the help of Altmann-Fitter, Liu (2009b) shows there is excellent fit for the modified right-truncated Zipf-Alekseev distribution to the Chinese dependency types. We also fit the distributions of Chinese dependency types with the Zipf-Alekseev distribution successfully ($R^2 = 0.995$). We further fit the distributions of ECSL learners’ dependency types with the Zipf-Alekseev distribution, as shown in Tables 3 and 4 and Figure 3. The fitting results for the rank-frequency distribution of dependency types across G1–G4 show a good fit with the Zipf-Alekseev distribution. Moreover, each (R^2) exceeds 0.95, indicating that L2 learners’ syntactic structures indeed follow the power-law distribution. This is additional evidence that there is a process with a diversified distribution. In response to Question 1, all the distributional features of dependency types in Chinese interlanguage also follow Zipf’s law like those in the natural languages.

Subjects (subj), *objects* (obj), *adverbials* (adva), *attributives* (atr), and *complement of usde’de’* (dec) were the most common dependency types, and their percentages always ranked in the top five. However, the construction *bei* (passiveness)(beis), *numeral adjuncts* (ma), and other dependency types accounted for a low percentage. Dependency types changed as Chinese proficiency improved, indicating that the dependency type tendencies among ECSL learners were changing. For example, the ratio of *subjects* to *objects* was unbalanced, and the percentage of *subjects* at G1

1 The *predicate complement* to which we refer is the special syntactic structure in Chinese. The complement in a Chinese sentence is all the additional elements (except the object) that follow the predicate, adjective or other predicate head. This is quite different from the complement in English.

2 *Subobject* and *subject complement* represents the pivotal construction.

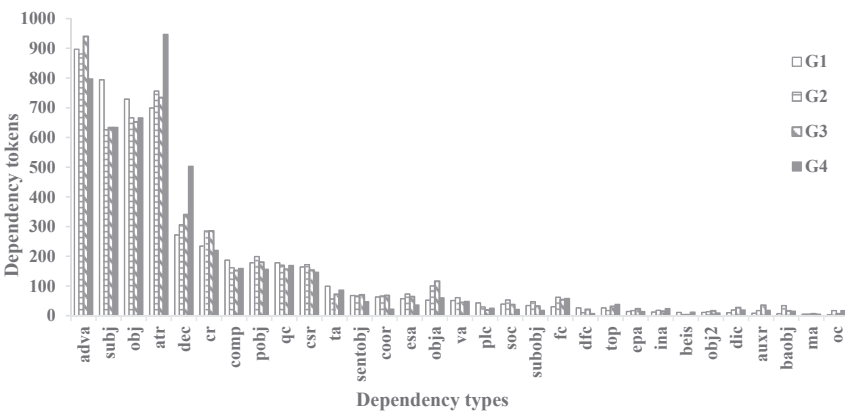


Figure 2: Distributions of overall dependency types across proficiency levels.

Table 3: Fitting the modified right-truncated Zipf-Alekseev distribution to dependency types.

Proficiency levels	χ^2	DF	P	N	C	T	R^2
G1	211.7167	27	0	32	0.0423	5,000	0.9553
G2	120.5779	26	0	31	0.0241	5,006	0.9752
G3	132.1599	26	0	31	0.0264	5,000	0.9674
G4	164.0481	26	0	31	0.0328	4,996	0.9667

χ^2 = chi-square; DF = degrees of freedom; P = Chi-square probability; C = discrepancy coefficient; N = the number of classes in the sample; T = the number of word tokens in the text; R^2 = determination coefficient.

(15.88%) was higher than that of any other proficiency levels, and also higher than the average of G1's objects (14.58%). However, ECSL learners produced more *objects* than *subjects* during G2–G4. This may have been because Chinese is a topic-prominent language, while English is a subject-prominent language (Chao 1968; Li and Thompson 1976). Chinese allows the ellipsis of the subjects (Lyu 1986; White 1989), but English is a non-null subject language (Hyams 1986). At the beginning of L2 learning, ECSL learners transfer the subject-prominence feature of native language English into second language (Jin 1994; Yuan 1995), and ECSL learners produce more overt subjects. However, with L2 proficiency improving, ECSL learners' syntactic features gradually approach those of the target language, and the percentage of subjects decreases. This demonstrates that advanced Chinese learners produced more sentences with null subjects than did beginners. We found that the total percentages of *attributives* (*atr*) and *adverbials* (*adva*) increased with improvement in Chinese proficiency, from 31.91% in G1 to 34.91% in G4. In the process of interlanguage development, the number of modifiers used by ECSL learners increased, while the percentages for clause adjunct and coordinating adjunct decreased from 4.68 to

Table 4: Fitting results for overall dependency types G1–G4.^a

G1				G2				G3				G4			
X[i]	F[i]	NP[i]		X[i]	F[i]	NP[i]		X[i]	F[i]	NP[i]		X[i]	F[i]	NP[i]	
adva	897	897		adva	881	881		adva	940	940		atr	947	947	
subj	794	944.81		atr	756	860.69		atr	734	734		adva	797	934.71	
obj	729	667.61		obj	666	631.38		obj	652	652		obj	666	671.99	
atr	699	485.21		subj	626	472.7		subj	633	459.03		subj	634	490.45	
dec	272	363.4		dec	306	362.9		dec	340	348.78		dec	503	366.83	
cr	234	279.31		cr	285	284.96		cr	285	272.45		cr	220	280.8	
comp	187	219.4		pobj	199	228.11		pobj	180	217.63		qc	169	219.35	
pobj	178	175.53		csr	172	185.61		qc	156	177.05		comp	159	174.37	
qc	178	142.62		qc	170	153.15		csr	153	146.26		pobj	156	140.71	
csr	164	117.44		comp	161	127.91		comp	152	122.41		csr	146	115.05	
ta	96	97.82		obja	100	107.95		obja	116	103.59		ta	86	95.16	
sentobj	68	82.32		esa	73	91.95		ta	72	88.52		obja	60	79.5	
coor	63	69.89		coor	66	78.97		sentobj	70	76.29		fc	58	67.02	
esa	57	59.81		sentobj	66	68.31		coor	68	66.24		va	48	56.95	
obja	52	51.56		fc	62	59.49		esa	64	57.91		sentobj	47	48.75	
va	51	44.73		ta	61	52.11		fc	54	50.93		top	38	42	
plc	43	39.03		ta	56	45.89		va	44	45.04		esa	36	36.41	
soc	39	34.24		soc	53	40.61		soc	37	40.03		plc	25	31.73	
subobj	34	30.19		subobj	47	36.11		auxr	35	35.74		ina	24	27.79	
fc	30	26.73		baobj	34	32.23		subobj	32	32.04		coor	22	24.45	
dfc	26	23.77		plc	29	28.88		top	31	28.83		soc	21	21.61	
top	26	21.22		dic	19	25.98		dic	27	26.04		dic	19	19.17	
epa	14	19.01		ina	18	23.44		epa	23	23.06		subobj	18	17.07	
ina	12	17.09		auxr	17	21.21		dfc	21	21.45		auxr	18	15.25	
beis	11	15.41		epa	17	19.26		plc	19	19.55		oc	17	13.67	

Table 4: (continued)

G1			G2			G3			G4		
$X[i]$	$F[i]$	$NP[i]$	$X[i]$	$F[i]$	$NP[i]$	$X[i]$	$F[i]$	$NP[i]$	$X[i]$	$F[i]$	$NP[i]$
obj2	11	13.93	oc	17	17.53	baobj	16	17.97	baobj	15	12.29
dic	10	12.63	top	17	15.99	obj2	16	16.37	epa	14	11.08
auxr	8	11.49	obj2	13	14.63	ina	15	15.04	beis	12	10.02
baobj	7	10.47	dfc	10	13.41	ma	6	13.84	obj2	9	9.08
ma	5	9.56	ma	5	12.32	oc	6	12.77	dfc	7	8.25
oc	3	8.75	beis	4	11.34	beis	3	11.8	ma	5	7.51
ua	2	8.03									

^aDue to space limitations, we only provide the fitting results examples for the overall dependency type distributions across G1–G4. $X[i]$ = observed dependency classes; $F[i]$ = observed dependency frequencies; $NP[i]$ = calculated frequency according to the modified right-truncated Zipf-Alekseev distribution; a , b , n and α are the parameters of the modified right-truncated Zipf-Alekseev distribution; subscripts 1, 2, 3, and 4 under each parameter represent ECSL learners' levels.

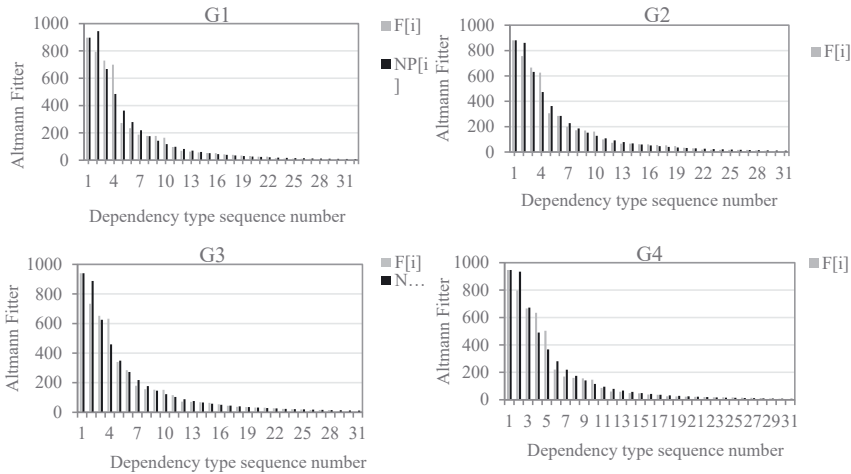


Figure 3: Fitting the modified right-truncated Zipf-Alekseev distribution to dependency type in G1–G4.

4.40% and 1.26 to 0.44%, respectively. In our dependency tagging system, chained clauses are also marked as *cr*. To an extent, the decreasing *cr* ratio reflects the decline in number of chained clauses produced by learners. Givón (2009) pointed out that the appearance and use of embedded structure and the subordinate component was an important sign of the increasing complexity of language syntactic structures. The development of vocabulary and syntax are likely intertwined, and difficult to disentangle (Jiang et al. 2019). Consequently, the increasing total percentage of attributives and adverbials and the decreasing total percentage of chained clauses and coordination likely represent the increased vocabulary knowledge and its deep vocabulary knowledge, such as syntactic features and collocation rules. In addition, *csr* stands for correlative adjunct in the clause adjunct. For example:

Yīnwèi wǒ zuótiān áoyè le , suǒyǐ wǒ jīntiān zǎoshàng shuǐlǎnjiào le
Because I yesterday stayed up late (AUX) , so I today morning oversleep (AUX)
‘Because I stayed up late last night, I slept in this morning.’

Áoyè ‘stay up late’ governs Yīnwèi ‘because’, the dependency relation between Áoyè and Yīnwèi is *csr*; shuǐlǎnjiào governs suǒyǐ, the dependency relation between shuǐlǎnjiào and suǒyǐ is *csr*; the dependency relation between the two governors (áoyè and shuǐlǎnjiào) is *cr*. Yīnwèi ... suǒyǐ ... is the representative collocation in Chinese compound sentences, representing the causal relationship. However, Chinese native speakers generally omit one or all of the conjunctions in authentic language communication. In the English sentence *Because I stayed up late last night, I slept in this morning*, the conjunction *because*, as far as we know, is indispensable.

By observing the changes in percentages of *cr* and *csr* together, we can comprehend the use of compound sentences and connectives in ECSL learners' Chinese interlanguage. At the outset, the ratio of compound sentences to connectives was 1.43. English is an overt cohesion-prominent, hypotactic language, and its clauses are often arranged with connectives. Meanwhile, Chinese emphasizes parataxis, and its words, phrases and clauses can be arranged sequentially without connectives between them, as long as the sentences are logical (Nida 1982). ECSL beginners have trouble comprehending this important typological feature in Chinese. Instead, they tend to overuse connectives. However, with improved language proficiency, ECSL learners can eliminate some connectives, and the number of connectives gradually decreases while the ratio of compound sentences to connectives increases.

Secondly, we investigated the distributions of governors' and dependents' classes across the four proficiency levels. According to the basic principle of dependency grammar, the distribution of dependents' classes can be regarded as the complete distribution of parts of speech (Liu 2009b). This is applicable to interlanguage. There were at least 15 governor classes and at least 18 dependent classes at each level of interlanguage. Table 5 shows the datum fitting with the Zipf-Alekseev distribution. In view of the R^2 for each proficiency level (each was above 0.97), this demonstrates that the distributions of governors' and dependents' classes fit the Zipf-Alekseev distribution. In accordance with the results of a study of German, Chinese, French, Latin and other languages (Liu 2009b; Strauss et al. 2007), interlanguage has a similar distribution model for parts of speech as natural languages, abiding by a power law distribution. For example, some classes are more common, such as nouns, verbs, adjectives, pronouns, and adverbs, whereas most word classes are less prevalent, such as conjunctions, prepositions and auxiliary words. Regardless of proficiency levels, verbs, nouns and adjectives are the three most common parts of

Table 5: Fitting the modified right-truncated Zipf-Alekseev distribution to a word class as the governor and dependent across proficiency levels.

Items	Proficiency levels	χ^2	DF	P	N	C	T^a	R^2
Governor classes	G1	38.4199	11	0.0001	16	0.0077	5,000	0.9994
	G2	68.9125	11	0	16	0.0138	5,006	0.9977
	G3	85.8124	12	0	17	0.0172	5,000	0.998
	G4	77.256	10	0	15	0.0155	4,996	0.9981
Dependent classes	G1	121.1034	9	0	14	0.0244	4,960	0.9793
	G2	90.5104	8	0	13	0.0184	4,927	0.9903
	G3	47.3429	7	0	12	0.0097	4,898	0.9892
	G4	187.7769	11	0	16	0.0378	4,966	0.9844

^aGovernor class *T* is inconsistent with the dependent classes. The reason for this discrepancy lies in the fact that L2 learners often made errors and produced incomplete sentences, leading to governors without dependents.

speech in interlanguage. Both the governor classes' and dependent classes' distribution regularities are diversified and uniform, similar to those of dependency types.

3.2 Dependency types' and dependent classes' distributions of verbs as governors

Previous studies have shown that the verb plays an important role in children's syntactic competence (Tomasello 1992). Moreover, the acquisition of verb functions is correlated with their syntactic competence (Bates et al. 1988). The verb is the center of the sentence and the dominant element. Moreover, verb is the most frequent POS tag for governors. Therefore, the study of verb development is the key to the study of syntactic development (Li and Liu 2019). The investigation of verbs' governable dependency types provides insight into verbs' syntactic variety. In other words, we can explore interlanguage verbs' syntactic functions, such as verb-object structures (*obj*) and adverbial structures (*adva*). Figure 4 and Table 6 show that the distribution of dependency types governable by a verb fit the Zipf-Alekseev distribution, with the goodness of fit R^2 ranging from 0.95 to 0.97. This demonstrates that the distribution of verb functions in Chinese interlanguage of varying proficiency levels follows the same pattern as natural languages.

Object (*obj*), *subject* (*subj*), *adverbial* (*adva*) and *clause adjunct* (*cr*) are the four most governable dependency types in interlanguage at all proficiency levels. This is consistent with the language of native Chinese speakers. The dependency type ratio

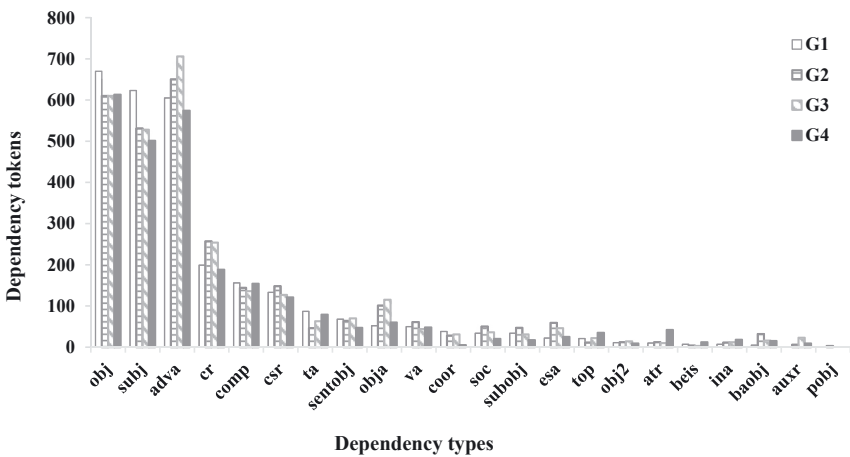


Figure 4: Changes in dependency types governable by a verb.

Table 6: Fitting the modified right-truncated Zipf-Alekseev distribution to the distribution of dependency types governable by a verb.

Proficiency levels	χ^2	DF	P	N	C	T	R^2
G1	124.7814	15	0	20	0.0441	2,832	0.9495
G2	85.1527	17	0	22	0.0295	2,887	0.9765
G3	67.6432	16	0	21	0.0233	2,897	0.9769
G4	69.5177	16	0	21	0.0268	2,592	0.9705

orders in interlanguage and target language are similar. Additionally, the dependency types that occur least often in L1 Chinese are also the ones that occur least often in interlanguage Chinese, such as *parenthetical expressions*, *adjunct of sentence end* (esa), and *topic* (top) (Gao et al. 2010). Although the distribution of dependency types governable by a verb (i.e., the verbs' syntactic functions) were unevenly distributed, ECSL learners as a whole displayed diversified patterns in their syntactic functions from the beginning of acquisition. There were 20 types of dependencies at G1, and advanced ECSL learners' verbs could govern 21 dependency types. This was different from Chinese children's native language acquisition, whose dependency types are governed by a verb at 14 months of age. According to Li and Liu (2019), 14 month-old Chinese children's verbs can only govern 5 dependency types. One important reason for this is the interaction between the new input and the first language representation (Ellis 2002). Unlike children's native language acquisition, adult L2 learners have a mature native linguistic system and an internalized cognitive system. Verbs are the center of both the grammatical structures and the semantic structures in a sentence. Adult ECSL learners' verb syntactic collocations are more diverse than those of Chinese children at the beginning of acquisition. Furthermore, adult L2 learners can produce more dependency types which demand high cognitive complexity, such as *adverbials* (adva), *clause adjuncts* (cr) and *aspect adjuncts* (ta).

As shown in Table 7, the distribution of the word classes governable by a verb fit the Zipf-Alekseev distribution, as Chinese natural language and the determination coefficients' R^2 are above 0.98. This accords with the law of diversified distribution of natural linguistic structures. The main POS governed by verbs in each proficiency level are noun, verb, pronoun and adverb. These four word classes alone account for over 10% of the total, while the others each account for less than 5%.

A word class's dependency properties represent its syntactic functions and syntactic combinations in a sentence. They also echo the distribution regularity of dependency types governable by a verb. The syntactic functions of verbs in Chinese

Table 7: Fitting the modified right-truncated Zipf-Alekseev distribution to a verb as the governor (dependent class word) across proficiency levels.

Proficiency levels	χ^2	DF	P	N	C	T	R^2
G1	57.1465	7	0	12	0.0206	2,770	0.9861
G2	76.1482	9	0	14	0.0265	2,874	0.9866
G3	64.8706	9	0	14	0.0225	2,883	0.9832
G4	39.2443	9	0	14	0.0153	2,569	0.9911

learners’ interlanguage are also diversified, and the typical and atypical dependents governed by verbs are distributed unevenly. The above discussion has provided a preliminary answer to Question 2, i.e., we explored the distribution of the main governor’s (verb) dependency properties in Chinese interlanguage, and obtained satisfactory results.

3.3 Dependency types’ and governor classes’ distributions of nouns as dependents

As is the case with Chinese native speakers (Liu 2009b), noun is the most frequent POS tag as a dependent in interlanguage (the percentages for noun used as a dependent were 32.18, 31.80, 26.64 and 32.29% in G1–G4 respectively, each ranking first in the dependents’ classes). Figure 5 shows the dependency type distribution, which can govern a noun. In view of the R^2 for each proficiency level (all of which were above 0.97), as shown in Table 8, the dependency types’ frequencies fit with a Zipf-Alekseev distribution and captured the power law distribution. We have demonstrated that the syntactic functions of Chinese interlanguage nouns conform to the diversification process from a mathematical perspective; dependency type diversity increased with increased language proficiency. There were 14 dependency types at G1, 16 dependency types at G2, 15 dependency types at G3 and 17 dependency types at G4 when nouns were dependents. Typical syntactic functions of Chinese nouns include *subject* (subj), *object* (obj) and *attribute* (atr), while non-typical syntactic functions include *predicate(s)*, *appositive* (epa) and *adverbial* (adva). Therefore, as shown in Figure 5, the percentages for *object*, *subject* and *attribute* are the most frequent, while the percentages for *appositive*, *adverbial*, and *predicate* are low. This shows that the dependency type distribution, which can govern a noun, also follows the unity of diversity and homogeneity.

In line with our expectations, the distribution of word classes (governors) which can govern a noun across proficiency levels also fits the Zipf-Alekseev distribution

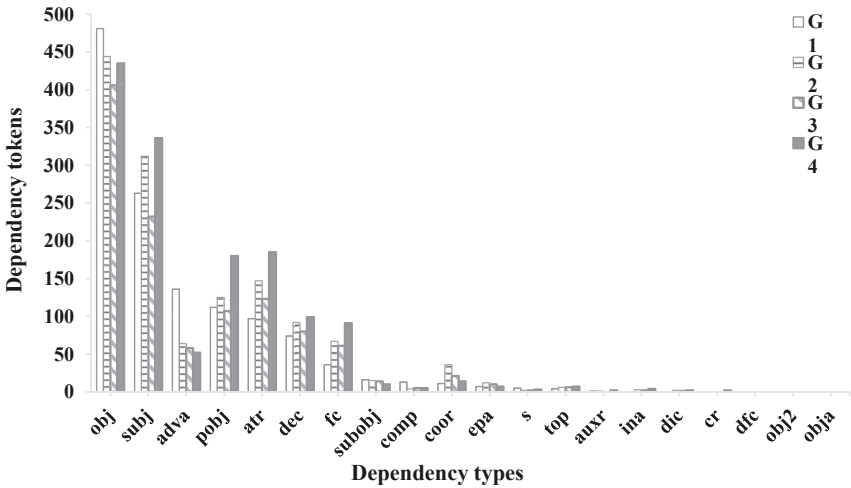


Figure 5: Changes in dependency types that govern a noun.

Table 8: Fitting the modified right-truncated Zipf-Alekseev distribution to a noun as the dependent (dependent type) across proficiency levels.

Proficiency levels	χ^2	DF	P	N	C	T	R^2
G1	31.04	5	0	10	0.0251	1,239	0.9891
G2	26.8642	5	0.0001	10	0.0204	1,314	0.9866
G3	21.8565	4	0.0002	9	0.0198	1,102	0.9886
G4	33.7738	3	0	8	0.0243	1,392	0.9785

($R^2 \geq 0.99$). This is shown in Table 9. Noun, verb and preposition across G1–G3 always rank in the top three, while noun, verb and conjunction rank in the top three at G4. So far, we can provide a complete answer to Question 2 based on the above analysis: the distribution of dependency properties to the governor and those to the dependent follow the same distributional regularities. This follows the laws of natural language.

Table 9: Fitting the modified right-truncated Zipf-Alekseev distribution to a noun as the dependent (word class of the governor) across proficiency levels.

Proficiency levels	χ^2	DF	P	N	C	T	R^2
G1	24.4277	3	0	8	0.0197	1,238	0.9982
G2	2.8664	2	0.2385	7	0.0022	1,305	0.9996
G3	16.1623	2	0.0003	7	0.0146	1,109	0.998
G4	7.0766	1	0.0078	6	0.005	1,413	0.9971

3.4 The evaluation dependency’s properties on L2 proficiency levels

Inspired by Zhao and Jiang (2020), we first explored Question 3 from a holistic perspective, fitting the rank-frequency distribution of dependency types, governors and dependents in each proficiency level to Zipf’s law. The detailed examination results in Table 10 show that the rank and the frequencies of the investigated interlanguage also followed the Zipf’s law distribution with high goodness of fit across four proficiency levels; parameter *a* decreased and parameter *b* increased with proficiency levels as shown in Figure 6. We verify Zhao and Jiang’s (2020) conclusion, providing additional statistical evidence that the parameters in Zipf’s law can reflect the Chinese interlanguage’s proficiency, to an extent. In addition, neither the value of parameters *a* or *b* in our study, nor those of Zhao and Jiang (2020), decreased or increased linearly. The development of parameters *a* and *b*, however, have fluctuations, and they develop in spurts at lower proficiency levels (G1–G2). Through the correlation analysis, we find a statistically significant correlation between the *a* values of Zipf’s law in overall dependents’ part-of-speech distribution and the proficiency levels ($r = -0.916, p < 0.05$). Additionally, there is a marginally significant positive correlation between proficiency levels and the *b* values of Zipf’s law in overall dependents’ class distribution ($r = 0.895, p = 0.052$). However, there is no significant correlation between parameters *a* or *b* for the overall dependency type distribution, the overall governors’ class distribution or proficiency levels ($ps > 0.05$), although *a* shows a descending trend and *b* shows an increasing trend. As a special natural language, one of interlanguage’s characteristics is repeatability. The development process from interlanguage to target language is not a linear forward progression. Rather, there are setbacks, twists and turns, and even retrogression. A complex and dynamic development process, it is full of nonlinear variation (Adjemian 1976; Larsen-Freeman 1997; Van Geert 2008). Therefore, the parameters which reflect L2 learners’ proficiency levels also fluctuate. Our results not only confirm the interlanguage theory, but also provide evidence for the interlanguage dynamic development theory.

Table 10: Fitting Zipf’s law of dependency properties to proficiency levels.

Items	G1	G2	G3	G4
Dependency type’s R^2	0.8681	0.8813	0.8811	0.9227
Governor class’s R^2	0.9294	0.9559	0.908	0.908
Dependent class’s R^2	0.8766	0.8746	0.9308	0.9465

In order to provide an in-depth answer to Question 3, and to verify the above results' credibility, we further fit Zipf's law to the distributions of verbs as governors and nouns as dependents. The fitting results of the distributions of dependency types governable by a verb and word classes governed by verbs are shown in Table 11 and Figure 7. They reveal a good fit with Zipf's law. Although the data fluctuated between G2 and G3, their overall fitting value R^2 increased with proficiency levels. In terms of the parameter's values, a and b fluctuated with Chinese proficiency. Through the correlation analysis, we find that there is a marginally significant positive correlation between the values of parameter b in the distribution of verbs' governable dependency types and proficiency levels ($r = 0.883$, $p = 0.058$). However, parameter a in the distribution of verbs' governable dependency types, as well as parameters a , b in the distribution of the word classes governable by a verb have no statistically significant correlation with proficiency levels ($ps > 0.05$). This reflects the dynamic features of interlanguage (Saville-Troike and Barto 2012).

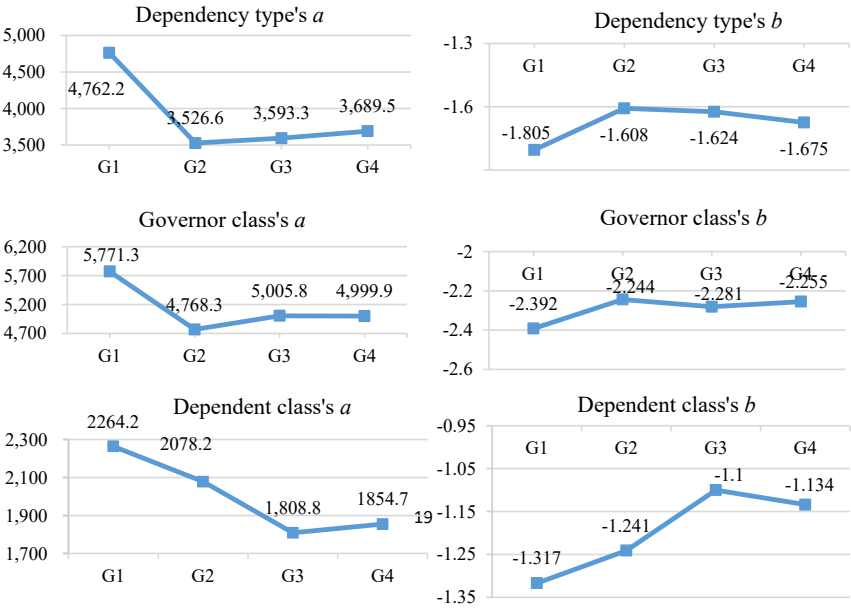


Figure 6: The overall change in parameters a and b with the development of proficiency levels.

Table 11: Fitting Zipf's law of dependency properties of the governor (verb) across proficiency levels.

Items	G1	G2	G3	G4
Dependency type's R^2	0.9076	0.8744	0.9043	0.9332
Dependent class's R^2	0.9026	0.8987	0.896	0.9289

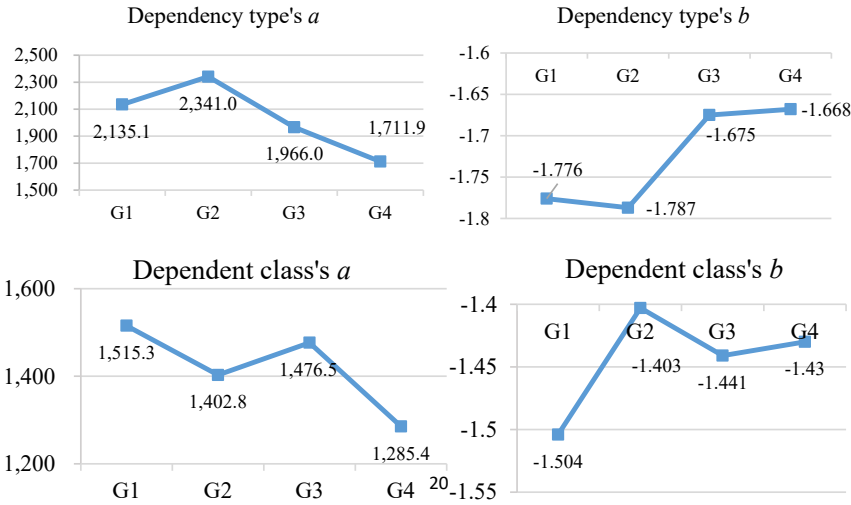


Figure 7: Change in parameters a and b with the increase in proficiency levels according to the dependent.

Regarding the distributions of the dependent's dependency properties, the distribution of the dependency types which can govern a noun, and that of word classes which can govern a noun, also fit Zipf's law. This is shown in Table 12. However, the former's goodness of fit was inferior to the latter's, especially at G4. The goodness of fit R^2 of the distribution of dependency types that can govern a noun is just above 0.75, but still within the acceptable range (Liu 2007). As shown in Figure 8, the parameter values a and b are also correlated with Chinese proficiency, and show a complex changing trend. Although the parameter b ascends in a straight line with increasing language proficiency, it does not increase at the same rate. For example, the value of b rises rapidly during G1–G3, and slowly in G3–G4. The values of the parameter b in the distribution of dependency types that govern a noun are significantly correlated with proficiency levels ($r = 0.946$, $p < 0.05$). The values of parameter b in the distribution of word classes (governors) which can govern a noun have a marginally significant correlation with proficiency levels ($r = 0.807$, $p = 0.096$). This demonstrates that there are peaks and troughs, advances and back sliding, and stabilization and attrition, in second language acquisition.

Table 12: Fitting Zipf's law of dependency properties of the dependent (noun) across proficiency levels.

Items	G1	G2	G3	G4
Dependency type's R^2	0.8865	0.8707	0.8791	0.7688
Governor class's R^2	0.904	0.9511	0.8839	0.9674

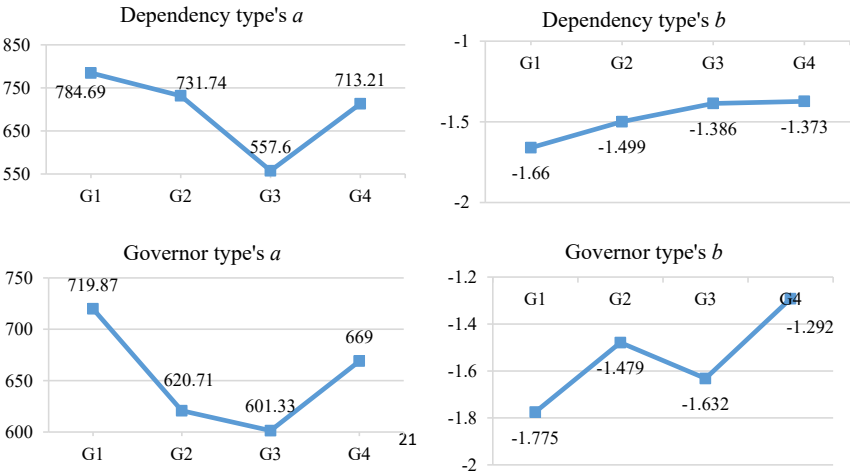


Figure 8: Change in parameters a and b with the increase in proficiency levels according to the governor.

4 Conclusions and implications

Existing studies of syntactic diversity in interlanguage have focused on changes in the frequency of partial and specific syntactic structures. In the present study, we selected the time-limited compositions of ECSL learners across four proficiency levels for syntactic tagging, based on dependency grammar. Using dependency properties as metrics, we investigated the syntactic diversity of Chinese interlanguage as a whole from the perspective of mathematical models.

In order to answer Questions 1 and 2, we have investigated the probability distributions of the dependency type, governor, dependent, verb as a governor, and noun as a dependent. Using the quantitative linguistics approach, we fit the datum with the Zipf-Alekseev distribution and Zipf's Law. In quantitative linguistics, Zipf's law and its variant, the Zipf-Alekseev model, represent the power law distribution of the linguistic units. This can characterize a diversification process. The results show that the investigated dependency properties of Chinese interlanguage can fit these two mathematical distribution models. Most of the corresponding goodness of fits are excellent, demonstrating that the power law distribution is universal, not only in natural language, but also in interlanguage and the general application of the laws of quantitative linguistics. In order to determine whether parameters a and b in Zipf's law are universal in measuring second language proficiency, i.e., to answer Question 3, we fit the distribution of dependency properties investigated with Zipf's law. We

found that parameters a and b in Zipf's law can reflect not only English, but also Chinese interlanguage's developmental proficiency trends. The values of a and b fluctuate with the increasing language proficiency. This demonstrates that Zipf's Law has the potential to serve as a universal metric for measuring interlanguage proficiency levels, and also reflects the dynamic features of interlanguage. In the future, more research and verification is needed on data from L2 learners across proficiency levels, NL backgrounds and TLs.

Our study demonstrates that interlanguage has the feature of diversification, which is consistent with human natural languages. Quantitative research on second language acquisition based on a corpus can objectively reflect L2 learners' linguistic use. According to the mathematical model, the interlanguage produced by the L2 has the features of both diversity and homogeneity, and it also follows the power law, which is in accordance with natural language. Adult L2 learners can produce various syntactic structures of the target language, and employ various syntactic functions of specific parts of speech. However, the typical dependency types are used frequently, and the atypical dependency types are less common. The same is true of the syntactic function of specific word classes, demonstrating that L2 learners also follow the principle of least effort in producing interlanguage.

This study is the first to adopt the quantitative linguistics approach and examine Chinese interlanguage as a whole from a holistic perspective. It reveals a unique law of interlanguage development and the universal linguistic law of human language development. Interlanguage is characterized by both diversity and uniformity, like human natural languages. L2 learners also follow the principle of least effort, and interlanguage displays the unique characteristics of being dynamic, gradual and repetitive. This paper validates and complements the existing theories of interlanguage research. It also applies quantitative linguistics methods developed by the Altmann School to the field of second language acquisition, expanding the research scope of quantitative linguistics. Previous studies suggest that writing prompts have an influence on measures of lexical and/or syntactic diversity (e.g., Alexopoulou et al. 2017; Way et al. 2000). Regrettably, the distribution of composition topics, the scores as well as the distribution of dependency properties in each writing were not precisely controlled or calculated in this paper. And we have not investigate the predictive effect of different granularity measures on writing quality of L2 learners across different proficiency levels. In the future, we will try to use the fine-grained linear regression analyses to investigate the development of syntactic diversity measures across different granularities.

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References

- Adjemian, Christian. 1976. On the nature of interlanguage systems. *Language Learning* 26(2). 297–320.
- Alexopoulou, Theodora, Marije Michel, Akira Murakami & Detmar Meurers. 2017. Task effects on linguistic complexity and accuracy: A large-scale learner corpus analysis employing natural language processing techniques. *Language Learning* 67(S1). 180–208.
- Altmann, Gabriel. 2005. Diversification processes. In Reinhard Köhler, Gabriel Altmann & Rajmund G. Piotrowski (eds.), *Quantitative linguistics. An international handbook*, 648–659. Berlin: de Gruyter.
- Bates, Elizabeth, Inge Bretherton & Lynn Sebestyen Snyder. 1988. *From first words to grammar: Individual differences and dissociable mechanisms*. Cambridge: Cambridge University Press.
- Best, Karl-Heinz. 2006. *Quantitative Linguistik. Eine Annäherung*. Göttingen: Peust & Gutschmidt.
- Bi, Peng & Jingyang Jiang. 2020. Syntactic complexity in assessing young adolescent EFL learners' writings: Syntactic elaboration and diversity. *System* 91. 102248.
- Bulté, Bram & Alex Housen. 2014. Conceptualizing and measuring short-term changes in L2 writing complexity. *Journal of Second Language Writing* 26. 42–65.
- Čech, Radek, Ján Mačutek, Zdeněk Žabokrtský & Aleš Horák. 2017. Polysemy and synonymy in syntactic dependency networks. *Digital Scholarship in the Humanities* 32(1). 36–49.
- Čech, Radek, Jiří Milička, Ján Mačutek, Michaela Koščová & Markéta Lopatková. 2018. Quantitative analysis of syntactic dependency in Czech. *Quantitative Analysis of Dependency Structures* 72. 53.
- Chao, Yuen Ren. 1968. *A grammar of spoken Chinese*. Berkeley and Los Angeles: University of California Press.
- Che, Wanxiang, Yunlong Feng, Libo Qin & Ting Liu. 2021. N-LTP: An open-source neural language technology platform for Chinese. In *Proceedings of the 2021 conference on empirical methods in natural language processing: System demonstrations*, 42–49. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics.
- Chen, Heng & Hai Xu. 2019. Quantitative linguistics approach to interlanguage development: A study based on the Guangwai-Lancaster Chinese learner corpus. *Lingua* 230. 102736.
- Corder, Stephen Pit. 1967. The significance of learner's errors. *International Review of Applied Linguistic* 5. 161–170.
- De Clercq, Bastien & Alex Housen. 2017. A cross-linguistic perspective on syntactic complexity in L2 development: Syntactic elaboration and diversity. *Modern Language Journal* 101(2). 315–334.
- Divjak, Dagmar. 2019. *Frequency in language: Memory, attention and learning*. Cambridge: Cambridge University Press.
- Durrant, Philip, Mark Brenchley & Lee McCallum. 2021. *Understanding development and proficiency in writing: Quantitative corpus linguistic approaches*. Cambridge: Cambridge University Press.
- Ellis, Nick C. 2002. Frequency effects in language processing: A review with implications for theories of implicit and explicit language acquisition. *Studies in Second Language Acquisition* 24(2). 143–188.
- Ellis, Nick C., Rita Simpson-Vlach, Ute Römer, Matthew Brook O'Donnell & Stefanie Wulff. 2015. Learner corpora and formulaic language in SLA. In Sylviane Granger, Gaëtanelle Gilquin & Fanny Meunier (eds.), *Cambridge handbook of learner corpus research*, 357–378. Cambridge, UK: Cambridge University Press.
- Ellis, Rod. 2003. *Task-based language learning and teaching*. Oxford: Oxford University Press.
- Ellis, Rod. 2009. The differential effects of three types of task planning on the fluency, complexity, and accuracy in L2 oral production. *Applied Linguistics* 30(4). 474–509.

- Eppler, Eva Duran. 2014. The dependency distance hypothesis for bilingual code-switching. In Kim Gerdes, Eva Hajičová & Leo Wanner (eds.), *Dependency linguistics: Recent advances in linguistic theory using dependency structures*, 183–206. Amsterdam, The Netherlands: John Benjamins.
- Gao, Song, Hongxin Zhang & Haitao Liu. 2014. Synergetic properties of Chinese verb valency. *Journal of Quantitative Linguistics* 21(1). 1–21.
- Gao, Song, Wei Yan & Haitao Liu. 2010. A quantitative study on syntactic functions of Chinese verbs based on dependency treebank. *Chinese Language Learning* 5. 105–112.
- Givón, Talmy. 2009. *The genesis of syntactic complexity: Diachrony, ontogeny, neuro-cognition, evolution*. Amsterdam, The Netherlands: John Benjamins Publishing.
- Gries, Stefan Th. 2008. Corpus-based methods in analyses of second language acquisition data. In Peter Robinson & Nick C. Ellis (eds.), *Handbook of cognitive linguistics and second language acquisition*, 406–431. New York/London: Routledge.
- Housen, Alex & Folkert Kuiken. 2009. Complexity, accuracy, and fluency in second language acquisition. *Applied Linguistics* 30(4). 461–473.
- Hřebíček, Luděk. 2002. Zipf's law and text. *Glottometrics* 3. 27–38.
- Hudson, Richard. 2007. *Language networks: The new word grammar*. Oxford: Oxford University Press.
- Hyams, Nina. 1986. *Language acquisition and the theory of parameters*. Dordrecht: Reidel.
- Jiang, Jingyang, Ouyang Jinghui & Liu Haitao. 2019. Interlanguage: A perspective of quantitative linguistic typology. *Language Sciences* 74. 85–97.
- Jin, Honggang. 1994. Topic-prominence and subject-prominence in L2 acquisition: Evidence of English-to-Chinese typological transfer. *Language Learning* 44(1). 101–122.
- Köhler, Reinhard. 2012. *Quantitative syntax analysis*, vol. 65. Berlin/Boston: Walter de Gruyter.
- Köhler, Reinhard, Altmann Gabriel & Rajmund G. Piotrowski. 2005. *Quantitative linguistics – An international handbook*. Berlin/NewYork: Mouton de Gruyter.
- Kyle, Kristopher. 2016. *Measuring syntactic development in L2 writing: Fine grained indices of syntactic complexity and usage-based indices of syntactic sophistication*. University of Georgia State PhD dissertation.
- Kyle, Kristopher & Scott A. Crossley. 2018. Measuring syntactic complexity in L2 writing using fine-grained clausal and phrasal indices. *The Modern Language Journal* 102(2). 333–349.
- Larsen-Freeman, Diane. 1997. Chaos/complexity science and second language acquisition. *Applied Linguistics* 18(2). 141–165.
- Li, Charles N. & Sandra A. Thompson. 1976. Subject and topic: A new typology of language. In Charles N. Li (ed.), *Subject and topic*, 457–489. Austin: University of Texas Press.
- Li, Hui & Haitao Liu. 2019. A quantitative study on the development of Chinese children's early verbal valence. *Applied Linguistics (Chinese)* 4. 131–140.
- Liu, Haitao. 2007. Probability distribution of dependency distance. *Glottometrics* 15. 1–12.
- Liu, Haitao. 2009a. *Dependency grammar: From theory to practice*. Beijing: Science Press.
- Liu, Haitao. 2009b. Probability distribution of dependencies based on a Chinese dependency treebank. *Journal of Quantitative Linguistics* 16(3). 256–273.
- Liu, Haitao. 2011. Quantitative properties of English verb valency. *Journal of Quantitative Linguistics* 18(3). 207–233.
- Liu, Haitao. 2017. *An introduction to quantitative linguistics*. Beijing: The Commercial Press.
- Liu, Haitao & Wei Huang. 2006. A Chinese dependency syntax for treebanking. In *Proceedings of the 20th Pacific Asia conference on language, information and computation*, 126–133. Huazhong Normal University, Wuhan, China: Tsinghua University Press.
- Liu, Haitao & Wei Huang. 2012. The current theory and method of quantitative linguistics. *Journal of Zhejiang University (Humanities and Social Sciences)* 42(02). 178–192.

- Lu, Xiaofei. 2011. A corpus-based evaluation of syntactic complexity measures as indices of college-level ESL writers' language development. *TESOL Quarterly* 45(1). 36–62.
- Lyu, Shuxiang. 1986. The flexibility of Chinese syntax. *Studies of the Chinese Language* 1. 1–9.
- Mel'čuk, Igor. 2003. Levels of dependency in linguistic description: Concepts and problems. In Ágel Vilmos, Ludwig Eichinger, Hans-Werner Eroms, Peter Hellwig, Hans Jürgen Heringer & Henning Lobin (eds), *Dependency and valency: An international handbook of contemporary research*, vol. 1, 188–229. Berlin – New York: W. de Gruyter.
- Meurers, Detmar & Markus Dickinson. 2017. Evidence and interpretation in language learning research: Opportunities for collaboration with computational linguistics. *Language Learning* 67(S1). 66–95.
- Mohanty, Panchanan & Ioan-Iovitz Popescu. 2014. Word length in Indian languages. *Glottometrics* 29. 95–109.
- Nemser, William. 1971. Approximative systems of foreign language learners. *IRAL-Interlanguage Review of Applied Linguistics in Language Learning* 9(2). 115–124.
- Nida, Eugene A. 1982. *Translating meaning*. San Dimas, CA: English Language Institute.
- Ninio, Anat. 2006. *Language and the learning curve: A new theory of syntactic development*. Oxford: Oxford University Press.
- Ortega, Lourdes. 2003. Syntactic complexity measures and their relationship to L2 proficiency: A research synthesis of college-level L2 writing. *Applied Linguistics* 24(4). 492–518.
- Ortega, Lourdes. 2012. Interlanguage complexity: A construct in search of theoretical renewal. In Bernd Kortmann & Benedikt Szmrecsanyi (eds.), *Linguistic complexity in interlanguage varieties, L2 varieties, and contact languages*, 127–155. Berlin: Walter de Gruyter.
- Ouyang, Jinghui & Jingyang Jiang. 2018. Can the probability distribution of dependency distance measure language proficiency of second language learners? *Journal of Quantitative Linguistics* 25(4). 295–313.
- Pan, Xiaxing & Haitao Liu. 2014. Adnominal constructions in modern Chinese and their distribution properties. *Glottometrics* 29. 1–30.
- Paquot, Magali. 2018. Phraseological competence: A missing component in university entrance language tests? Insights from a study of EFL learners' use of statistical collocations. *Language Assessment Quarterly* 15(1). 29–43.
- Park, Ji-Hyun. 2017. *Syntactic complexity as a predictor of second language writing proficiency and writing quality*. Ann Arbor, MI: Michigan State University Unpublished PhD Dissertation.
- Popescu, Ioan-Iovitz, Karl-Heinz Best & Gabriel Altmann. 2014. *Unified modeling of length in language (= studies in quantitative linguistics 16)*. Lüdenscheid: RAM-Verlag.
- Saville-Troike, Muriel & Karen Barto. 2012. *Introducing second language acquisition*. London: Cambridge University Press.
- Selinker, Larry. 1972. Interlanguage. *International Review of Applied Linguistics* 10(3). 219–231.
- Strauss, Udo, Peter Grzybek & Gabriel Altmann. 2007. Word length and word frequency. In Peter Grzybek (ed.), *Contributions to the science of text and language*, 277–294. Dordrecht: Springer.
- Tesnière, Lucien. 1959. *Éléments de la syntaxe structurale*. Paris: Klincksieck.
- Tomasello, Michael. 1992. *First verbs: A case study of early grammatical development*. Cambridge: C. U. P.
- Van Geert, Paul. 2008. The dynamic systems approach in the study of L1 and L2 acquisition: An introduction. *Modern Language Journal* 92(2). 179–199.
- Way, Denise Paige, Elizabeth G. Joiner & Michael A. Seaman. 2000. Writing in the secondary foreign language classroom: The effects of prompts and tasks on novice learners of French. *The Modern Language Journal* 84(2). 171–184.
- White, Lydia. 1989. *Universal grammar and second language acquisition*. Amsterdam: John Benjamins.
- Yih, Tsy. 2021. Review of Divjak (2019): Frequency in language: Memory, attention and learning. *Review of Cognitive Linguistics* 19(2). 596–601.

- Yuan, Boping. 1995. Acquisition of base-generated topics by English-Speaking learners of Chinese. *Language Learning* 45(4). 567–603.
- Yue, Ming & Haitao Liu. 2011. Probability distribution of discourse relations based on a Chinese RST-annotated corpus. *Journal of Quantitative Linguistics* 18(2). 107–121.
- Zhang, Hongxi & Haitao Liu. 2017. Motifs of generalized valencies. In Haitao Liu & Junying Liang (eds.), *Motifs in language and text*, 231–260. Berlin/Boston: De Gruyter Mouton.
- Zhao, Qianying & Jingyang Jiang. 2020. Verb valency in interlanguage: An extension to valency theory and new perspective on L2 learning. *Poznań Studies in Contemporary Linguistics* 56(2). 339–363.
- Zhu, Dexi. 1985. *The questions and answers on grammar*. Beijing: Beijing Commercial Press.
- Zipf, George Kingsley. 1935. *The psycho-biology of language: An introduction to dynamic philology*. London: George Routledge & Sons Ltd.
- Zipf, George Kingsley. 1949. *Human behaviour and principle of least effort*. Cambridge: Addison-Wesley.